Proof by Mathematical Induction

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The three steps of induction**

State the three steps required for a valid proof by mathematical induction.

(1) Base case: verify the statement holds for the smallest value of n (usually $n = 1$). (2) Inductive hypothesis: assume the statement holds for $n = k$. (3) Inductive step: use the assumption to prove the statement holds for $n = k + 1$. Concluding, by the principle of mathematical induction, the statement holds for all n in the given range.

2 **[Foundational]** • **Target: Proving a summation formula**

Prove by induction that $1 + 2 + 3 + \cdots + n = n(n + 1)/2$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 1$. RHS $= 1(2)/2 = 1$. True. Inductive hypothesis: assume $1 + 2 + \cdots + k = k(k + 1)/2$. For $n = k + 1$: $1 + 2 + \cdots + k + (k + 1) = k(k + 1)/2 + (k + 1)$. Factoring out $(k + 1)$: this is $(k + 1)(k/2 + 1) = (k + 1)(k + 2)/2$, which matches the formula with $n = k + 1$. Since it holds for $n = 1$ and $n = k \Rightarrow n = k + 1$, by induction it holds for all $n \geq 1$.

3 **[Foundational]** • **Target: Proving divisibility results**

Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$.

Base case ($n = 1$): $7^1 - 1 = 6$, divisible by 6. Assume $7^k - 1 = 6m$ for some integer m, so $7^k = 6m + 1$. For $n = k + 1$: $7^{k+1} - 1 = 7 \cdot 7^k - 1 = 7(6m + 1) - 1 = 42m + 6 = 6(7m + 1)$, which is divisible by 6. By induction, $7^n - 1$ is divisible by 6 for all $n \geq 1$.

4 **[Foundational]** • **Target: Induction with inequalities**

Prove by induction that $2^n > n^2$ for all integers $n \geq 5$.

Base case ($n = 5$): $2^5 = 32$, $5^2 = 25$, and $32 > 25$. True. Assume $2^k > k^2$ for some $k \geq 5$. For $n = k + 1$: $2^{k+1} = 2 \cdot 2^k > 2k^2$. We need $2k^2 \geq (k + 1)^2 = k^2 + 2k + 1$, i.e. $k^2 - 2k - 1 \geq 0$, which holds for $k \geq 3$ (true since $k \geq 5$). So $2^{k+1} > 2k^2 \geq (k + 1)^2$. By induction, $2^n > n^2$ for all $n \geq 5$.

5 **[Foundational]** • **Target: Induction in calculus: derivatives of a product**

Prove by induction that $\frac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}$ for all positive integers n.

Base case ($n = 1$): differentiating e^{ax} once gives $ae^{ax} = a^1 e^{ax}$, matching the formula. Inductive hypothesis: assume the k-th derivative of e^{ax} is $a^k e^{ax}$. Differentiating this once more: $\frac{d}{dx}(a^k e^{ax}) = a^k \cdot ae^{ax} = a^{k+1} e^{ax}$ — this is exactly the $(k + 1)$-th derivative, matching the formula at $n = k + 1$. By induction, true for all positive integers n.

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6 [Foundational] • Target: Induction with a recursively defined sequence

A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n - 1$. Prove by induction that $u_n = 2^n + 1$ for all $n \geq 1$.

Base case ($n = 1$): $u_1 = 3$ (given). Formula: $2^1 + 1 = 3$. True. Assume $u_k = 2^k + 1$. Then $u_{k+1} = 2u_k - 1 = 2(2^k + 1) - 1 = 2^{k+1} + 2 - 1 = 2^{k+1} + 1$, matching the formula at $n = k + 1$. By induction, $u_n = 2^n + 1$ for all $n \geq 1$.

7 [Foundational] • Target: Proving a geometric-series-style summation

Prove by induction that the sum $3^0 + 3^1 + \cdots + 3^{n-1}$ equals $(3^n - 1)/2$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 3^0 = 1$. RHS $= (3^1 - 1)/2 = 2/2 = 1$. True. Assume the sum $3^0 + \cdots + 3^{k-1} = (3^k - 1)/2$. For $n = k + 1$: adding the next term 3^k gives $(3^k - 1)/2 + 3^k$. Combining over a denominator of 2: this is $(3^k - 1 + 2 \cdot 3^k)/2 = (3 \cdot 3^k - 1)/2 = (3^{k+1} - 1)/2$, matching the formula at $n = k + 1$. By induction, true for all $n \geq 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Proving a divisibility result with two variables

Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers $n \geq 1$.

Base case ($n = 1$): $1^3 + 2(1) = 3$, divisible by 3. Assume $k^3 + 2k = 3m$ for some integer m. For $n = k + 1$: $(k + 1)^3 + 2(k + 1) = k^3 + 3k^2 + 3k + 1 + 2k + 2$. Grouping terms: this is $(k^3 + 2k) + (3k^2 + 3k + 3) = 3m + 3(k^2 + k + 1)$. Factoring out 3: $3m + 3(k^2 + k + 1) = 3(m + k^2 + k + 1)$, which is divisible by 3. By induction, $n^3 + 2n$ is divisible by 3 for all $n \geq 1$.

9 [Intermediate] • Target: Proving a divisibility result with two variables

Prove by induction that $9^n - 1$ is divisible by 8 for all positive integers $n \geq 1$, and explain briefly why a similar argument would show $a^n - 1$ is divisible by $a - 1$ for any integer $a > 1$.

Base case ($n = 1$): $9^1 - 1 = 8$, divisible by 8. Assume $9^k - 1 = 8m$ for some integer m, so $9^k = 8m + 1$. For $n = k + 1$: $9^{k+1} - 1 = 9 \cdot 9^k - 1 = 9(8m + 1) - 1 = 72m + 9 - 1 = 72m + 8 = 8(9m + 1)$, divisible by 8. By induction, true for all $n \geq 1$. The same inductive pattern works for any base a: $a^{k+1} - 1 = a \cdot a^k - 1 = a[(a - 1)m' + 1] - 1 = (a - 1)(am') + (a - 1) = (a - 1)(am' + 1)$, always a multiple of $a - 1$.

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10 **[Intermediate]** • **Target: Induction with a recurrence relation for a sequence of sums**

A sequence satisfies $S_1 = 2$ and $S_{n+1} = S_n + 6n + 2$ for $n \geq 1$. Prove by induction that $S_n = 3n^2 - n$ for all $n \geq 1$.

Base case ($n = 1$): $S_1 = 2$ (given). Formula: $3(1)^2 - 1 = 2$. True. Assume $S_k = 3k^2 - k$. Then $S_{k+1} = S_k + 6k + 2 = 3k^2 - k + 6k + 2 = 3k^2 + 5k + 2$. The formula at $n = k + 1$ gives $3(k+1)^2 - (k+1) = 3k^2 + 6k + 3 - k - 1 = 3k^2 + 5k + 2$, which matches exactly. Since it holds for $n = 1$ and $n = k \Rightarrow n = k + 1$, by induction $S_n = 3n^2 - n$ for all $n \geq 1$.

11 **[Intermediate]** • **Target: A closing check: identifying an invalid induction argument**

A student claims to prove that all horses are the same color using induction on the number of horses n in a group: the base case $n = 1$ is trivially true (one horse is trivially "the same color" as itself), and the student then argues that if any k horses are the same color, then any $k + 1$ horses must also be the same color. Explain precisely where this argument breaks down, and state why a correct induction proof must avoid this flaw.

The flaw is in the inductive step: the argument tries to overlap two groups of k horses (the first k and the last k out of $k + 1$ total) and conclude they share a common color through the overlap. This overlap argument only works when the overlap is non-empty, which requires $k \geq 2$. It fails silently at the transition from $n = 1$ to $n = 2$, since two groups of 1 horse each (drawn from a total of 2) have no horse in common, so there is no shared reference point linking their colors. This illustrates why a c.

12 **[Intermediate]** • **Target: Proving a divisibility result by induction**

Prove by induction that $n^3 - n$ is divisible by 6 for all integers $n \geq 1$.

Base case $n = 1$: $1^3 - 1 = 0$, and 0 is divisible by 6. Inductive step: assume $m^3 - m = 6k$ for some integer k. Then $(m+1)^3 - (m+1) = m^3 + 3m^2 + 3m + 1 - m - 1$. Simplifying: this is $m^3 + 3m^2 + 2m$. Regrouping: this is $(m^3 - m) + 3m(m+1) = 6k + 3m(m+1)$. Since m and $m + 1$ are consecutive integers, one of them is even, so $m(m+1)$ is even and $3m(m+1)$ is divisible by 6. Thus $(m+1)^3 - (m+1) = 6k + 6j = 6(k + j)$ for some integer j, which is divisible by 6. By induction, $n^3 - n$ is divisible by 6 for all $n \geq 1$.

13 **[Intermediate]** • **Target: Induction in combinatorics: counting subsets**

Prove by induction that a set with n elements has exactly 2^n subsets, for all integers $n \geq 1$.

Base case $n = 1$: a 1-element set $\{a\}$ has exactly 2 subsets ($\varnothing$ and $\{a\}$), and $2^1 = 2$. True. Inductive step: assume every k-element set has 2^k subsets. Consider a $(k+1)$-element set formed by adding one new element x to a k-element set. Every subset of the $(k+1)$-element set either excludes x (giving exactly the 2^k subsets of the original k-element set) or includes x (giving one more subset for each of those same 2^k subsets, with x added in). This gives.

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14 **[Intermediate]** • **Target: A closing check: identifying an invalid induction argument**

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n\to\infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

(a) Lower bound via AM-GM: For any positive real number $x_n > 0$, the Arithmetic Mean-Geometric Mean inequality states that $\frac{a+b}{2} \geq \sqrt{ab}$. With $a = x_n$ and $b = \frac{5}{x_n}$: $x_{n+1} = \frac{x_n + 5/x_n}{2} \geq \sqrt{x_n \cdot \frac{5}{x_n}} = \sqrt{5}$. Thus $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Monotonicity & Convergence: Compute the difference: $x_{n+1} - x_n = \frac{1}{2}x_n + \frac{5}{2x_n} - x_n = \frac{5 - x_n^2}{2x_n}$. Since $x_n \geq \sqrt{5}$ for $n \geq 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Proving a sum-of-products formula by induction** • **Cross-Lesson: 6.01 Matrices (AI HL)**

Express $z = -1 + i\sqrt{3}$ in polar form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$, with principal argument $-\pi < \theta \leq \pi$. Hence find z^6 and determine all three solutions to $w^3 = z$.

1. Modulus: $r = |z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$. Argument: since z lies in Quadrant II, $\theta = \pi - \arctan(\sqrt{3}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$. Polar form: $z = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$; Exponential form: $z = 2e^{i2\pi/3}$. 2. Compute z^6 via De Moivre's Theorem: $z^6 = 2^6 e^{i\cdot 6(2\pi/3)} = 64e^{i4\pi} = 64(\cos 4\pi + i\sin 4\pi) = 64(1+0i) = 64$. 3. Solve $w^3 = z = 2e^{i2\pi/3}$.

16 **[Challenging]** • **Target: Proving an exponential inequality by induction** • **Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)**

Prove by mathematical induction that $5^n - 1$ is divisible by 4 for all positive integers $n \in \mathbb{Z}^+$.

Base Step: For $n = 1$: $5^1 - 1 = 4 = 4(1)$. Since 4 is divisible by 4, the statement $P(1)$ is true. Inductive Hypothesis: Assume $P(k)$ is true for some positive integer $k \geq 1$, meaning:

$5^k - 1 = 4m$ for some integer $m \in \mathbb{Z} \Rightarrow 5^k = 4m + 1$. Inductive Step: Consider $n = k + 1$:

$5^{k+1} - 1 = 5 \cdot 5^k - 1$. Substitute the inductive hypothesis $5^k = 4m + 1$:

$5(4m+1) - 1 = 20m + 5 - 1 = 20m + 4 = 4(5m+1)$. Since $m \in \mathbb{Z}$, $5m + 1$ is an integer, so.

17 **[Challenging]** • **Target: Proving a divisibility result by induction** • **Cross-Lesson: 6.03 Markov Chains (AI HL)**

Solve the equation $z^3 + 8 = 0$, expressing all solutions in Cartesian form $a + bi$.

Write $z^3 = -8 = 8e^{i\pi}$. By De Moivre's Theorem, the three cube roots are: $z_k = 2\exp\left(i\frac{\pi + 2\pi k}{3}\right)$ for $k = 0, 1, 2$. For $k = 0$: $z_0 = 2e^{i\pi/3} = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 1 + i\sqrt{3}$. For $k = 1$: $z_1 = 2e^{i\pi} = 2(-1) = -2$. For $k = 2$: $z_2 = 2e^{i5\pi/3} = 2e^{-i\pi/3} = 2\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = 1 - i\sqrt{3}$. Notice that $1 \pm i\sqrt{3}$ are complex conjugates, as predicted by the Conjugate Root Theor.

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18 [Challenging] • Target: Proving a closed form for a recursive sequence • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

[Algebra, Proof & Complex Numbers · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

19 [Challenging] • Target: A second exponential inequality by induction • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Use the infinite sum formula $S_\infty = \frac{a_1}{1-r}$: $25 = \frac{a_1}{1-(-0.4)} = \frac{a_1}{1.4} \Rightarrow a_1 = 25 \times 1.4 = 35$. Compute the 4th partial sum $S_4 = \frac{a_1(1-r^4)}{1-r}$: $(-0.4)^4 = 0.0256$. $S_4 = \frac{35(1-0.0256)}{1.4} = \frac{35(0.9744)}{1.4} = 25 \times 0.9744 = 24.36$.

20 [Challenging] • Target: A third inequality by induction • Cross-Lesson: 1.04 Partial Fractions

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x \, dx$ for integers $n \geq 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \geq 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x \, dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n\to\infty} I_n$.

(a) Base integral: $I_0 = \int_0^1 e^x \, dx = [e^x]_0^1 = e^1 - e^0 = e - 1$. (b) Reduction formula via IBP: Let $u = x^n \Rightarrow du = nx^{n-1} \, dx$. Let $dv = e^x \, dx \Rightarrow v = e^x$. $I_n = [x^n e^x]_0^1 - n\int_0^1 x^{n-1} e^x \, dx = (1^n e^1 - 0) - nI_{n-1} = e - nI_{n-1}$. ■ (c) Compute I_3: - $I_1 = e - 1(I_0) = e - (e-1) = 1$. - $I_2 = e - 2(I_1) = e - 2(1) = e - 2$. - $I_3 = e - 3(I_2) = e - 3(e-2) = e - 3e + 6 = 6 - 2e$. (d) Squeeze Theorem & Limit: For.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The three steps of induction

In proving by induction that $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$, assuming the result is true for $n = k$, what must be added to $\frac{k(k+1)}{2}$ to obtain the sum for $n = k + 1$?

a. k

b. $k + 2$

c. $k + 1$ ✓

d. $2k + 1$

The $(k+1)$th term of the sum is $k+1$, so we add $k+1$ to reach $\frac{(k+1)(k+2)}{2}$.

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22 [Foundational] • Target: Proving a summation formula

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers n. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.

a. **100** ✓

b. 110

c. 10

d. 210

Proof by Induction: Base Case ($n = 1$): $a_1 = 1$. The formula gives $1^2 = 1$. Thus, the base case holds. Inductive Hypothesis: Assume $a_k = k^2$ for some positive integer k. Inductive Step: We need to show that $a_{k+1} = (k+1)^2$. From the recurrence relation, $a_{k+1} = a_k + 2k + 1$. Using the inductive hypothesis, $a_{k+1} = k^2 + 2k + 1 = (k+1)^2$. Thus, by induction, $a_n = n^2$ for all positive integers n. Closed-form expression for S_N: $S_N = \sum_{n=1}^{N} a_n = \sum_{n=1}^{N} n^2$. This is the su.

23 [Foundational] • Target: Proving divisibility results

Using the first three terms of the Maclaurin series for e^x, approximate $e^{1 \times \frac{1}{4}}$ as a fraction.

a. $\frac{173}{160}$

b. $\frac{41}{32}$ ✓

c. $\frac{253}{160}$

d. $\frac{237}{160}$

$e^x \approx 1 + x + \frac{(x)^2}{2}$, at $x = \frac{1}{4}$: $1 + \frac{1}{4} + \frac{\frac{1}{16}}{2} = \frac{41}{32}$

24 [Foundational] • Target: Induction with inequalities

For $x^3 - 19x - 30 = 0$, find the product of the three roots.

a. **30** ✓

b. 43

c. 27

d. 20

Working through this complex-numbers step: Product of roots $= -\frac{d}{a} = -(-30) = 30$

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25 [Foundational] • Target: Induction in calculus: derivatives of a product

The first, fourth, and sixth terms of an arithmetic progression (u_n) are equal to the first, second, and third terms, respectively, of an infinite geometric progression (v_n) with common ratio $re1$. Given that the sum to infinity of the geometric progression is 27, determine the maximum value of $S_n = \sum_{k=1}^{n} u_k$, the sum of the first n terms of the arithmetic progression.

a. 44

b. 45 ✓

c. 50

d. 54

Let the arithmetic progression (u_n) have first term a and common difference d. Then: $u_1 = a, \quad u_4 = a + 3d, \quad u_6 = a + 5d.$ Let the geometric progression (v_n) have first term v_1 and common ratio r. Since $u_1 = v_1$, $u_4 = v_2$, and $u_6 = v_3$, we have: $v_1 = a, \quad v_2 = ar, \quad v_3 = ar^2.$ Equating terms: 1) $a + 3d = ar \Rightarrow 3d = a(r-1) \Rightarrow d = \frac{a(r-1)}{3}$. 2) $a + 5d = ar^2$. Substitute d into the second equation.

26 [Foundational] • Target: Induction with a recursively defined sequence

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. **5** ✓

c. 6

d. 7

The sum of the first n terms of a geometric sequence is given by the formula $S_n = a_1 \frac{r^n - 1}{r - 1}$, where a_1 is the first term and r is the common ratio. Given $a_1 = 5$ and $r = 2$, and $S_n = 315$, we can substitute these values into the formula: $315 = 5\frac{2^n - 1}{2 - 1}$ $315 = 5\frac{2^n - 1}{1}$ $315 = 5(2^n - 1)$ Divide both sides by 5: $\frac{315}{5} = 2^n - 1$ $63 = 2^n - 1$ Add 1 to both sides: $63 + 1 = 2^n$ $64 = 2^n$ Since $2^6 = 64$, we have $n = 6$. Let's check the choices.

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27 [Foundational] • Target: Proving a geometric-series-style summation

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. Determine the sum of the x-intercepts.

a. **4** ✓

b. 3

c. 1

d. -3

The x-intercepts of a function are the values of x for which $f(x) = 0$. For the quadratic function $f(x) = x^2 - 4x + 3$, we can find the x-intercepts by solving the equation $x^2 - 4x + 3 = 0$. This is a quadratic equation that can be factored. We are looking for two numbers that multiply to 3 and add to -4. These numbers are -1 and -3. Therefore, the equation can be factored as $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. These are the x-intercepts. The sum of the x-intercepts is $1 + 3 = 4$. Alternative.

28 [Foundational] • Target: Proving an inequality involving factorials

A sequence is defined by the recurrence relation $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. If $a_1 = 2$, find the value of a_{2023}.

a. **2** ✓

b. 0

c. -2

d. 1

Let's compute the first few terms of the sequence: $a_1 = 2$ $a_2 = a_1^2 - 2 = 2^2 - 2 = 4 - 2 = 2$ $a_3 = a_2^2 - 2 = 2^2 - 2 = 4 - 2 = 2$ It appears that for $a_1 = 2$, all subsequent terms of the sequence are also 2. We can prove this by induction. Base case: For $n = 1$, $a_1 = 2$, which is true. Inductive step: Assume that for some integer $k \geq 1$, $a_k = 2$. We need to show that $a_{k+1} = 2$. From the recurrence relation, $a_{k+1} = a_k^2 - 2$. By the inductive hypothesis, $a_k = 2$. Substituting this into.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Proving a divisibility result with two variables

Product of roots of $2x^2 - 5x + 3 = 0$:

a. $\frac{3}{2}$ ✓

b. $\frac{5}{2}$

c. $-\frac{3}{2}$

d. 3

Applying Algebra, Proof & Complex Numbers principles, Product = c/a = 3/2. Therefore, the verified correct option is $\frac{3}{2}$.

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30 [Intermediate] • Target: Induction with a recurrence relation for a sequence of sums

For $x^2 - x - 56 = 0$, find the product of the roots.

a. -42

b. -47

c. **-56** ✓

d. -51

Working through this complex-numbers step: Product of roots $= \frac{c}{a} = -56 = -56$

31 [Intermediate] • Target: Induction with a strengthened hypothesis

Simplify $(1 + 2i)(5 + 5i)$, giving your answer in the form $x + yi$.

a. $-3 + 16i$

b. $-5 + 15i$ ✓

c. $-1 + 12i$

d. $-3 + 14i$

Working through this complex-numbers step: $(1 + 2i)(5 + 5i) = (5) + (5)i + 2(5)i + 2(5)i^2 = -5 + 15i$

32 [Intermediate] • Target: Induction to prove a formula for the sum of cubes

Write $z = 4(\cos 180^\circ + i\sin 180^\circ)$ in the form $x + yi$ (exact values).

a. $-4 + 1$

b. 4

c. -4 ✓

d. $-4i$

Working through this complex-numbers step: $z = 4\cos 180^\circ + 4\sin 180^\circ\, i = -4$

33 [Intermediate] • Target: Induction to prove a formula for a weighted geometric sum

The first three terms of a geometric progression are given by $u_1 = 3k - 2, \quad u_2 = k + 2, \quad u_3 = k - 2,$ where k is a real constant. Given that the infinite geometric series $\sum_{n=1}^{\infty} u_n$ converges, find the sum to infinity, S_∞.

a. 32 ✓

b. -1

c. $\frac{32}{3}$

d. 16

Step 1: Set up the common ratio property of a geometric sequence. Since the terms form a geometric progression, the ratio between consecutive terms must be equal: $\frac{u_2}{u_1} = \frac{u_3}{u_2} \Rightarrow u_2^2 = u_1 u_3$ Step 2: Substitute the expressions in terms of k: $(k + 2)^2 = (3k - 2)(k - 2)$ $k^2 + 4k + 4 = 3k^2 - 8k + 4$ Step 3: Solve the quadratic equation for k: $2k^2 - 12k = 0 \Rightarrow 2k(k - 6) = 0$ Thus, $k = 0$ or $k = 6$. Step 4: Apply the convergence criterion $|r| < 1$: - If $k = 0$.

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34 [Intermediate] • Target: A closing check: identifying an invalid induction argument

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n - 1)$ ✓

b. $a_n = 3n + 2$

c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

The formula for the n-th term of an arithmetic sequence is given by $a_n = a_1 + d(n - 1)$, where a_1 is the first term and d is the common difference. In this case, $a_1 = 5$ and $d = 3$. Substituting these values into the formula gives $a_n = 5 + 3(n - 1)$. If we simplify this, we get $a_n = 5 + 3n - 3 = 3n + 2$. Therefore, the direct formula is $a_n = 5 + 3(n - 1)$ or its simplified form $a_n = 3n + 2$. Choice A directly uses the definition. Choice B is the simplified form. Let's examine all options: Choice A.

35 [Intermediate] • Target: Proving a sum-of-products formula by induction

Consider the quadratic function $f(x) = x^2 - 8x + k$. The graph of $y = f(x)$ is a parabola that intersects the x-axis at the points $(2, 0)$ and $(6, 0)$. Find the value of k.

a. **12** ✓

b. 16

c. 20

d. 24

Since the parabola intersects the x-axis at $(2, 0)$ and $(6, 0)$, these are the roots of the quadratic equation $f(x) = 0$. Therefore, we can write the function in factored form as $f(x) = a(x - 2)(x - 6)$. Expanding this, we get $f(x) = a(x^2 - 8x + 12)$. Comparing this with the given function $f(x) = x^2 - 8x + k$, we can see that the coefficient of x^2 is 1. Thus, $a = 1$. So, $f(x) = x^2 - 8x + 12$. By comparing the constant terms, we find that $k = 12$. Alternatively, we can substitute one of the roots into the o.

36 [Intermediate] • Target: Proving an exponential inequality by induction

The first three terms of a geometric progression are 32, 24, and 18. Find the sum to infinity, S_∞, of the progression.

a. 128 ✓

b. 96

c. 74

d. 64

The first term of the geometric progression is $a = 32$. The common ratio r is found by dividing any term by the preceding term: $r = \frac{24}{32} = \frac{3}{4}$ Since $|r| = \left|\frac{3}{4}\right| < 1$, the geometric progression converges and the sum to infinity exists. The sum to infinity is given by: $S_\infty = \frac{a}{1 - r}$ Substituting $a = 32$ and $r = \frac{3}{4}$: $S_\infty = \frac{32}{1 - \frac{3}{4}} = \frac{32}{\frac{1}{4}} = 32 \times 4 = 128$ Common misconceptions: - Choice B (96) arises fro.

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37 [Intermediate] • Target: Proving a divisibility result by induction

The diagram shows the graph of the function $y = f(x)$. The function is of the form $f(x) = x^2 + bx + c$. The graph passes through the point $(2, -3)$ and has a minimum value of -4 at $x = 1$. Find the value of b.

a. -2

b. 1

c. **2** ✓

d. -1

The function is given by $f(x) = x^2 + bx + c$. The minimum value of a quadratic function in the form $ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. In this case, $a = 1$, so the minimum occurs at $x = -\frac{b}{2}$. We are given that the minimum occurs at $x = 1$. Therefore, $1 = -\frac{b}{2}$, which implies $b = -2$. Alternatively, we know the vertex is at $(1, -4)$. The vertex form of a quadratic is $f(x) = a(x - h)^2 + k$. Given $a = 1$ and the vertex $(h, k) = (1, -4)$, we have $f(x) = 1(x - 1)^2 - 4$. Expanding thi.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Proving a closed form for a recursive sequence • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

The graph of the function $g(x) = x^2 - 6x + k$ is shown below. The graph intersects the x-axis at points P and Q. Point P has coordinates $(2, 0)$. Find the value of the parameter k.

a. 1

b. **2** ✓

c. 4

d. 8

The graph of the quadratic function $g(x) = x^2 - 6x + k$ intersects the x-axis at points where $g(x) = 0$. We are given that one of these intersection points is P, with coordinates $(2, 0)$. This means that when $x = 2$, $g(x) = 0$. We can substitute these values into the function to find the value of k: $g(2) = (2)^2 - 6(2) + k$ $0 = 4 - 12 + k$ $0 = -8 + k$ $k = 8$ Therefore, the value of the parameter k is 8.

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39 [Challenging] • Target: A second exponential inequality by induction • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Consider the quadratic function $f(x) = x^2 - 4x + 3$. Find the value of x where the graph of $f(x)$ intersects the x-axis at the point $(x, 0)$, given that $x > 1$.

a. 2

b. **3** ✓

c. 4

d. 5

The graph of $f(x)$ intersects the x-axis when $f(x) = 0$. We set the function equal to zero and solve for x: $x^2 - 4x + 3 = 0$ Factoring the quadratic equation, we get: $(x - 1)(x - 3) = 0$ This gives two possible solutions for x: $x = 1$ or $x = 3$. The question specifies that we need to find the value of x where $x > 1$. Therefore, the correct value is $x = 3$.

40 [Challenging] • Target: A third inequality by induction • Cross-Lesson: 1.04 Partial Fractions

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points where $f(x) = 0$. Determine the sum of the x-coordinates of these intersection points.

a. 1

b. 3

c. **4** ✓

d. -3

The x-intercepts of the graph of $y = f(x)$ are the roots of the equation $f(x) = 0$. For a quadratic equation of the form $ax^2 + bx + c = 0$, the sum of the roots is given by $-\frac{b}{a}$. In this case, $f(x) = x^2 - 4x + 3$, so $a = 1$, $b = -4$, and $c = 3$. The sum of the x-coordinates of the intersection points is $-\frac{-4}{1} = 4$.

41 [Challenging] • Target: Induction in combinatorics: counting subsets • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

A geometric sequence has first term $a = 1$ and common ratio $r = 2$. Find the sum of the first 5 terms.

a. **31** ✓

b. -31

c. 32

d. 16

$S_n = \frac{a(r^n - 1)}{r - 1} \Rightarrow S_5 = \frac{((2)^5 - 1)}{2 - 1} = 31.$

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42 [Challenging] • Target: Proving the sum of the first n odd numbers • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Arithmetic series sum: S_{10} for $1 + 3 + 5 + \cdots$ (first 10 odd numbers):

a. **100** ✓

b. 55

c. 50

d. 99

Sum of first n odd numbers $= n^2 = 100$.

43 [Challenging] • Target: The three steps of induction • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Consider the quadratic equation $x^2 - kx + 8 = 0$, where k is a real parameter. The graph of the corresponding quadratic function $y = x^2 - kx + 8$ is a parabola. Given that the parabola intersects the x-axis at exactly one point, find the value of k.

a. $-4\sqrt{2}$

b. $-2\sqrt{2}$

c. $2\sqrt{2}$

d. $4\sqrt{2}$ ✓

The quadratic equation $ax^2 + bx + c = 0$ has exactly one real root (i.e., the parabola $y = ax^2 + bx + c$ intersects the x-axis at exactly one point) if and only if its discriminant, $\Delta = b^2 - 4ac$, is equal to zero. In the given equation, $x^2 - kx + 8 = 0$, we have $a = 1$, $b = -k$, and $c = 8$. Therefore, the discriminant is $\Delta = (-k)^2 - 4(1)(8) = k^2 - 32$. For the parabola to intersect the x-axis at exactly one point, we must have $\Delta = 0$. So, $k^2 - 32 = 0$. This gives $k^2 = 32$, which m.

44 [Challenging] • Target: Proving a summation formula • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Let z_1 and z_2 be complex numbers such that $z_1 = 2 + i$ and $z_2 = 3 - 2i$. ; Consider the complex number $w = \frac{z_1^2}{z_2}$. ; If $w = a + bi$, where a and b are real numbers, find the value of $a^2 + b^2$.

a. $\frac{25}{13}$ ✓

b. $\frac{5}{13}$

c. $\frac{169}{25}$

d. $\frac{13}{5}$

First, we calculate z_1^2: $z_1^2 = (2 + i)^2 = 4 + 4i + i^2 = 4 + 4i - 1 = 3 + 4i$. ; Next, we calculate $w = \frac{z_1^2}{z_2} = \frac{3 + 4i}{3 - 2i}$. ; To simplify, we multiply the numerator and denominator by the conjugate of the denominator: $w = \frac{(3 + 4i)(3 + 2i)}{(3 - 2i)(3 + 2i)} = \frac{9 + 6i + 12i + 8i^2}{9 - 4i^2} = \frac{9 + 18i - 8}{9 + 4} = \frac{1 + 18i}{13} = \frac{1}{13} + \frac{18}{13}i$. ; So, $a = \frac{1}{13}$ and $b = \frac{18}{13}$. ; We need to find $a^2 + b^2$.

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45 [Challenging] • Target: Proving divisibility results • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Consider the complex number $w = 4 - 3i$. Find the modulus of w, denoted as $|w|$.

a. **5** ✓

b. 6

c. 7

d. 25

The modulus of a complex number $z = x + yi$ is given by $|z| = \sqrt{x^2 + y^2}$. For the complex number $w = 4 - 3i$, we have $x = 4$ and $y = -3$. Therefore, $|w| = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$. Common errors include: - Forgetting to square the imaginary part: $\sqrt{4 + (-3)} = \sqrt{1} = 1$ (Incorrect). - Forgetting to take the square root: $4^2 + (-3)^2 = 16 + 9 = 25$ (Incorrect, as this is the square of the modulus). - Incorrectly handling the negative sign in the imaginary part.

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